

Application Note

Design guide for DDR-SDRAM & Clock source

User Manual

MDVR3511FAN200601-V1

Ver. 1.0 / February 3, 2006

VR3511F

*Application Note for DDR-SDRAM & Clock source of
VR3511F Processor*

MAGIC EYES

This document contains information on a product under development. MagicEyes reserves the right to change or discontinue this product without notice. © MagicEyes Digital Co.ltd 2005. All rights reserved.

Important Notice

Information in this document is provided solely to enable system and software implementers to use MagicEyes products. There are no express or implied copyright licenses granted hereunder to design or fabricate any integrated circuits or integrated circuits based on the information in this document.

MagicEyes reserves the right to make changes without further notice to any products herein.

MagicEyes makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MagicEyes assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation consequential or incidental damages. "Typical" parameters which may be provided in MagicEyes data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. MagicEyes does not convey any license under its patent rights nor the rights of others. MagicEyes products are not designed, intended, or authorized for use as components in systems intended for surgical implant into the body, or other applications intended to support or sustain life, or for any other application in which the failure of the MagicEyes product could create a situation where personal injury or death may occur. Should Buyer purchase or use MagicEyes products for any such unintended or unauthorized application, Buyer shall indemnify and hold MagicEyes and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that MagicEyes was negligent regarding the design or manufacture of the part.

© MagicEyes Digital

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electric or mechanical, by photocopying, recording, or otherwise, without the prior written consent of MagicEyes Digital.

HOW TO REACH US:

MagicEyes Digital co., Ltd.

463-824 #501, A-ONE Plaza, 269-6 Seohyeon-Dong, Bundang-Gu
Seongnam-City, Gyeonggi-Do, Korea
TEL: 82-31-788-0300
FAX:82-31-707-5765

HOME PAGE:

<http://www.mesdigital.com>

DEVELOPER FORUM:

<http://forum.mesdigital.com>

Table of Contents

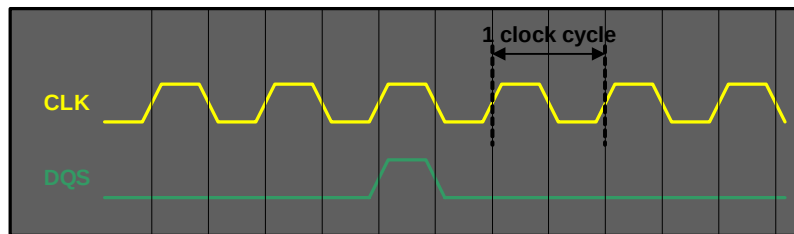
1. Design guide for stable DDR-SDRAM operation.....	4
1.1. Circuit design guide	4
1.2. PCB Layout design guide	9
1.2.1. PCB layout rule related to DDR-SDRAM.....	9
2. Design guide for Clock source.....	12
2.1. Circuit design guide	12
2.2. PCB layout design guide	12
2.2.1. PCB layout rule related to clock source.....	12

1. Design guide for stable DDR-SDRAM operation

Unlike SDR-SDRAM, DDR-SDRAM fetches data at rising and falling edge of clock signal, so the timing margin to write data is much poor than SDR-SDRAM. Therefore much care is necessary during schematic design and PCB layout process. This bulletin describes the circuit design & PCB layout guide for stable DDR-SDRAM operation of VR3511F product.

1.1. Circuit design guide

<Fig 1> shows the DDR-SDRAM Memory Write/Read timing diagram for general memory controller case.

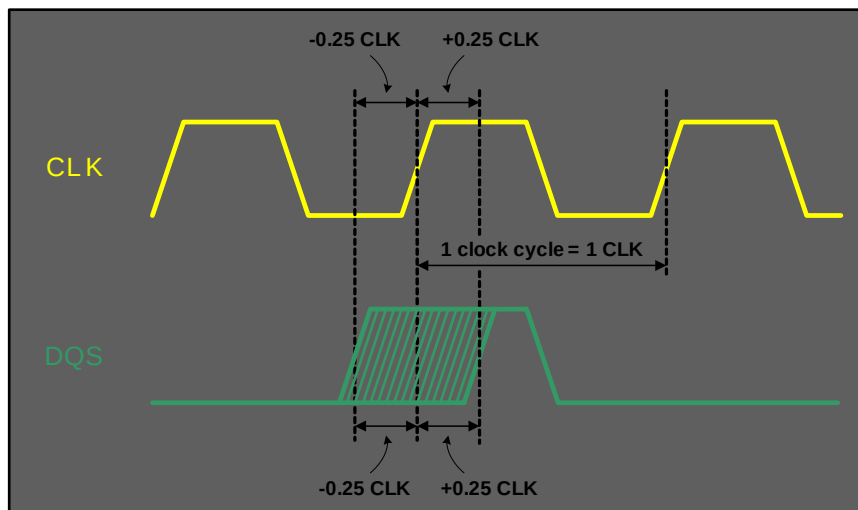


<Fig 1. General timing diagram of CLK and DQS of DDR-SDRAM>

In Memory Write case the memory controller writes data to DDR-SDRAM and DQS is generated by the memory controller. In Memory Read case the memory controller reads data from DDR-SDRAM and unlike DDR-SDRAM Memory Write case, DQS is generated by DDR-SDRAM. You don't need to worry about the DDR-SDRAM Memory Read case because the DQS signal is automatically generated by DDR-SDRAM. But much care is required in the DDR-SDRAM Memory Write case because the DQS signal is generated by the memory controller, and the timing relationship between CLK and DQS must follow DDR-SDRAM timing specification.

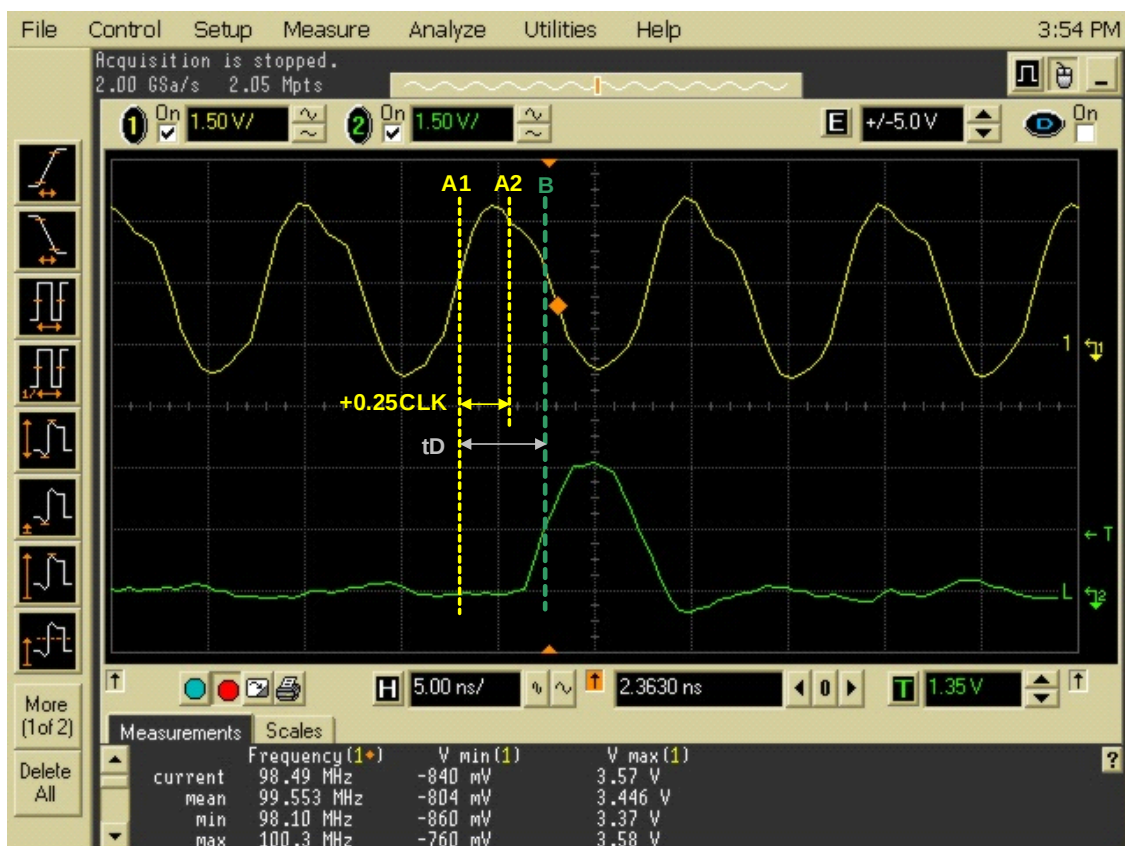
<Fig 2> shows ideal timing margin between CLK and DQS. The rising edge of DQS is located between -0.25CLK and $+0.25\text{CLK}$ of the rising edge of CLK. However the actual DDR-SDRAM product sold in market has more margin than $\pm 0.25\text{CLK}$

Note) The timing margin between CLK and DQS is not fixed to $\pm 0.25\text{CLK}$. Some of DDR-SDRAM has the timing margin of $\pm 0.28\text{CLK}$.



<Fig 2. Ideal timing margin of CLK and DQS>

<Fig 3> shows CLK and DQS signal timing diagram when VR3511F writes data to DDR-SDRAM (Memory Write case). **YELLOW**-colored signal is CLK and **GREEN**-colored signal is DQS. It is an image-capture file of oscilloscope screen.



<Fig 3. Memory Write timing diagram of CLK and DQS for VR3511F case>

Parameter definition of <Fig 3>

A1 : the rising edge of CLK

A1 ~ A2 : the timing margin between CLK and DQS for stable Memory Write operation (+0.25CLK)

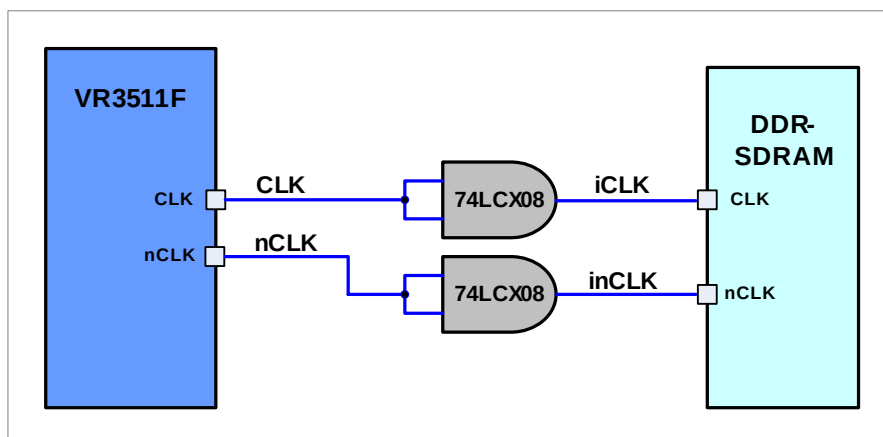
B : the rising edge of DQS

tD : the distance between **A1** and **B**

You can see the fact that DQS rising edge (**B**) is out of +0.25CLK range of CLK rising edge (**A1**), that is, the distance of **tD** is more longer than the distance between **A1** and **A2**(+0.25CLK), and as a result DDR-SDRAM Memory Write is unstable (There is no need to consider for DDR-SDRAM Memory Read case because the DQS is automatically generated by DDR-SDRAM in DDR-SDRAM Memory Read case).

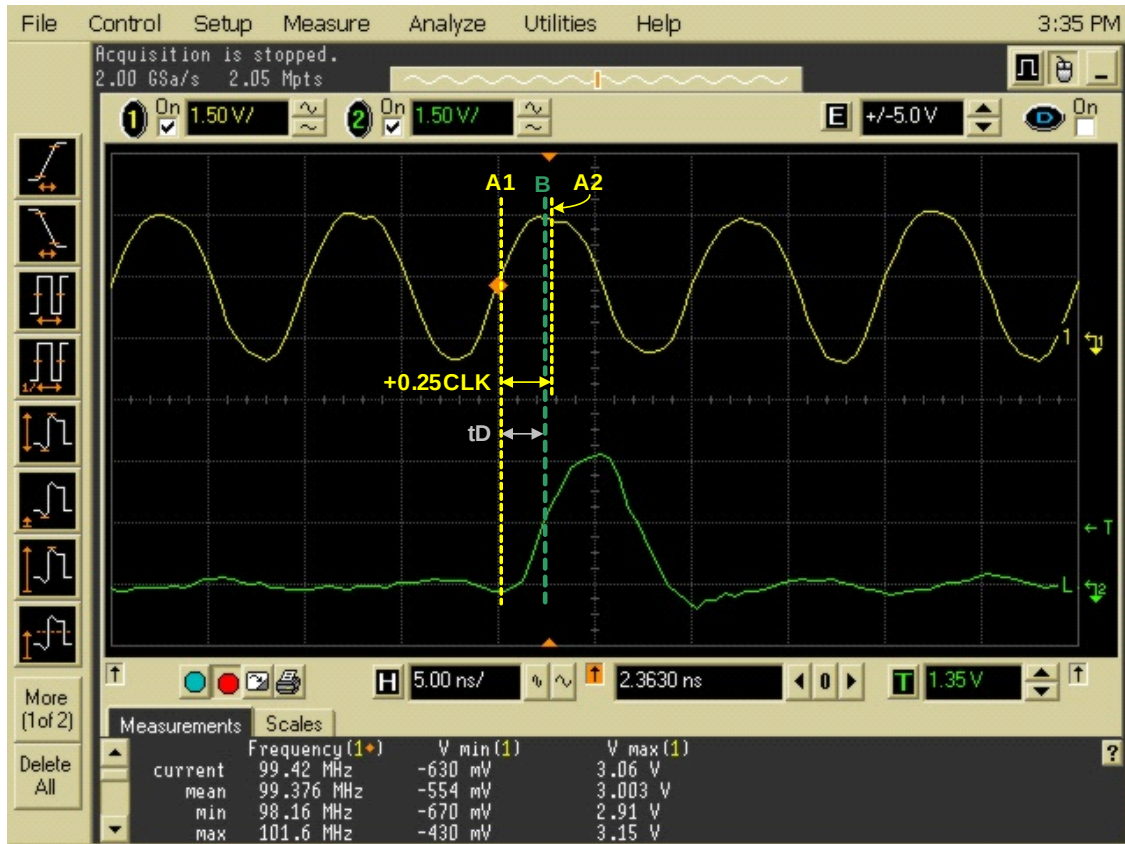
To make a stable operation in DDR-SDRAM Memory Write case, CLK signal must be shifted toward DQS signal and this CLK shifting is achieved by using the delay-IC. There are lots of delay-IC products in IC market and many kinds of ICs can be used as delay-IC however the most important things are cost & stability issue. MagicEyes tested some of delay-ICs for CLK shifting and choose 74LCX08 because of the advantage of cost & stability. Originally 74LCX08 is high speed AND-Gate IC. The propagation delay of 74LCX08 is 2.2 ~ 2.4ns in 2.5V Voltage case and this delay makes stable timing margin between CLK and DQS.

<Fig 4> shows the DDR-SDRAM interface that delay-IC(74LCX08) is inserted between VR3511F and DDR-SDRAM.



<Fig 4. Block diagram to shift CLK signal>

<Fig 5> shows the timing diagram of CLK and DQS in this delay-IC case.



<Fig 5. Timing diagram of CLK and DQS in one gate delay case>

Parameter definition of <Fig 5>

A1 : the rising edge of CLK

A1 ~ A2 : the timing margin between CLK and DQS for stable Memory Write operation (+0.25CLK)

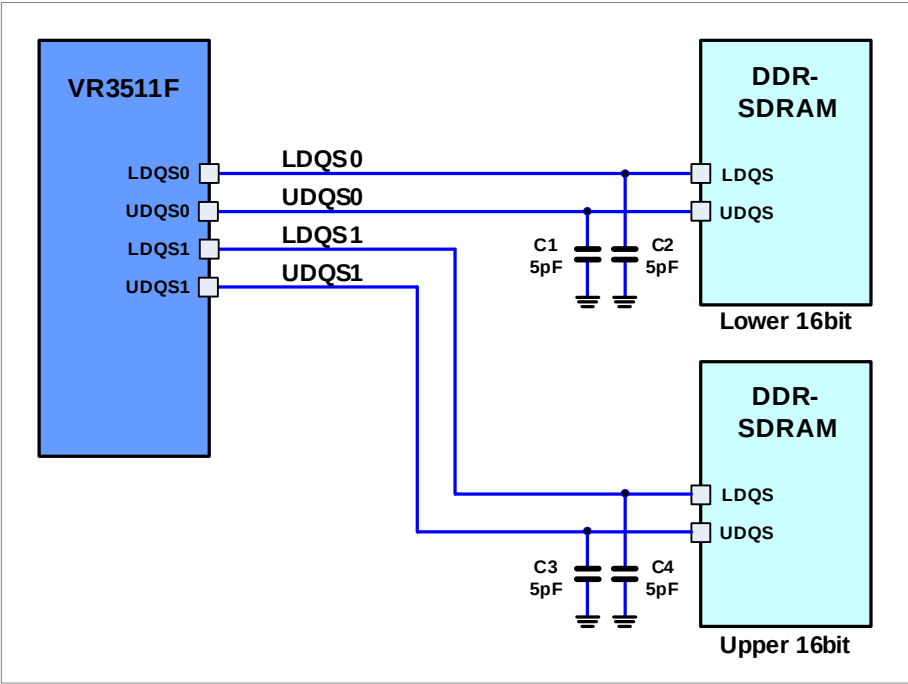
B : the rising edge of DQS

tD : the distance between **A1** and **B**

Compared with <Fig 3>, CLK is shifted about 2.3ns toward DQS and the location of DQS rising edge(**B**) is inside +0.25CLK range(**A1 ~ A2**) so Memory Write operation is accomplished stably.

From the circuit design point of view, the insertion of 74LCX08 between VR3511F and DDR-SDRAM is a solution for stable Memory operation.

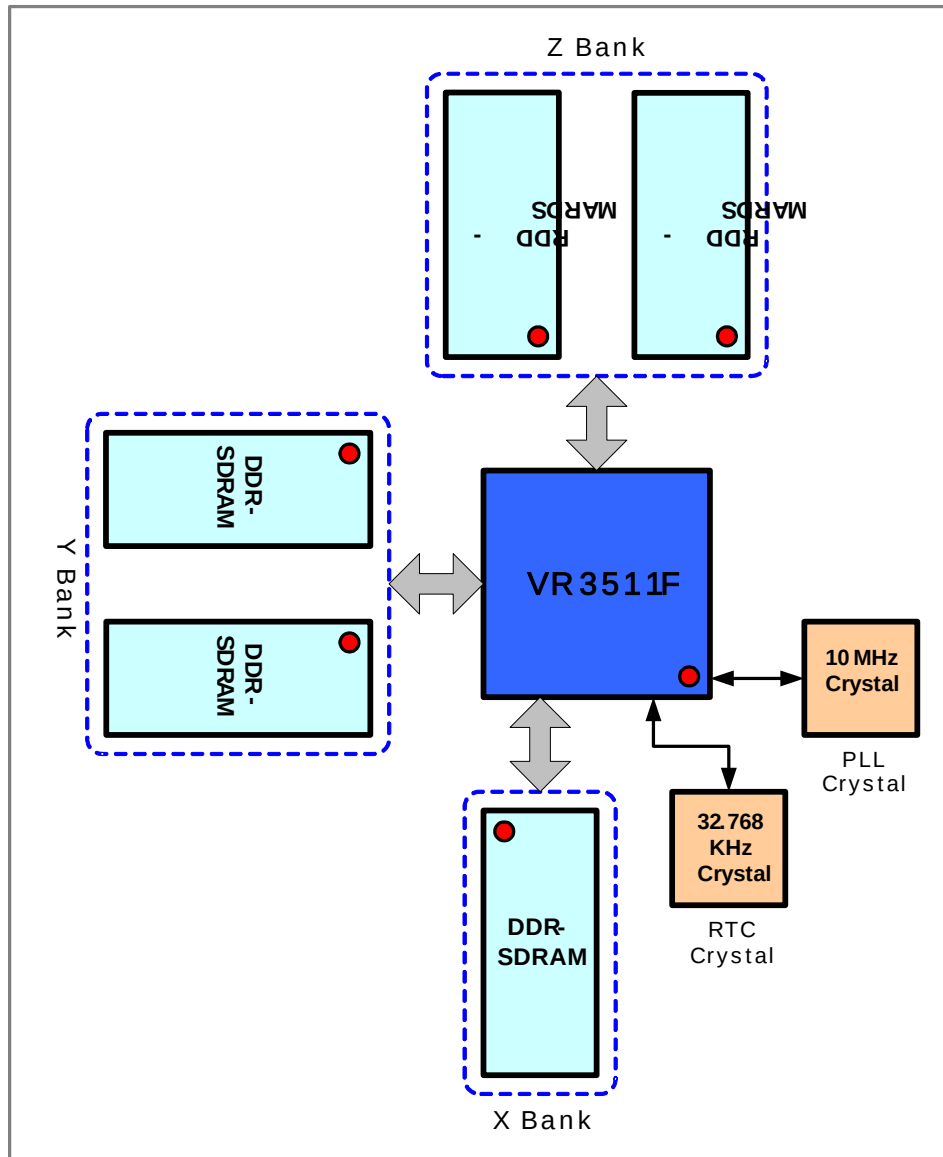
In addition to the insertion of 74LCX08, all DQS pins of DDR-SDRAM must be connected to 5pF capacitor. Refer to <Fig 6>.



<Fig 6. Capacitor connection of DQS pin of DDR-SDRAM>

1.2. PCB Layout design guide

PCB layout about DDR-SDRAM is required special attention because it needs many fastidious restrictions. This PCB Layout guide describes the DDR-SDRAM location, each signal's pattern width, each signal's pattern length, via hole count, and so on. The customer of VR3511F must follow this bulletin's PCB Layout guide.



<Fig 7. The major component arrangement related to VR3511F>

1.2.1. PCB layout rule related to DDR-SDRAM

- 01) Regard the location of DDR-SDRAM & two crystal as the most important thing

VR3511F Application Note

- 02) VR3511F and DDR-SDRAM must be located on same plane of PCB
- 03) The location and direction of each DDR-SDRAM must be same as <Fig 7>
- 04) The order of priority on pattern routing process :
DQ(Data bus) → CLK(nCLK) & DQS → DM → CKE,nCS,nRAS,nCAS,nWE → Address
- 05) The length of all DQ(Data bus) line of DDR-SDRAM must be same
- 06) Do not use via hole(through hole) on all DQ(Data bus) line
- 07) The length of all DQS line of DDR-SDRAM must be same
- 08) The length of DQS line must be longer than the length of DQ line
- 09) C1, C2, C3, and C4 of <Fig 6> must be located nearby DQS pin of DDR-SDRAM as close as possible
- 10) The length of CLK and nCLK must be same (refer to <Fig 3>)
- 11) The length of iCLK and inCLK must be same (refer to <Fig 3>)
- 12) The length of (CLK+iCLK) line must be longer than the length of DQ line

**Recommended values for each signal's length and width of DDR-SDRAM are as follows.
The customer of VR3511F must follow <Table 1>'s values.**

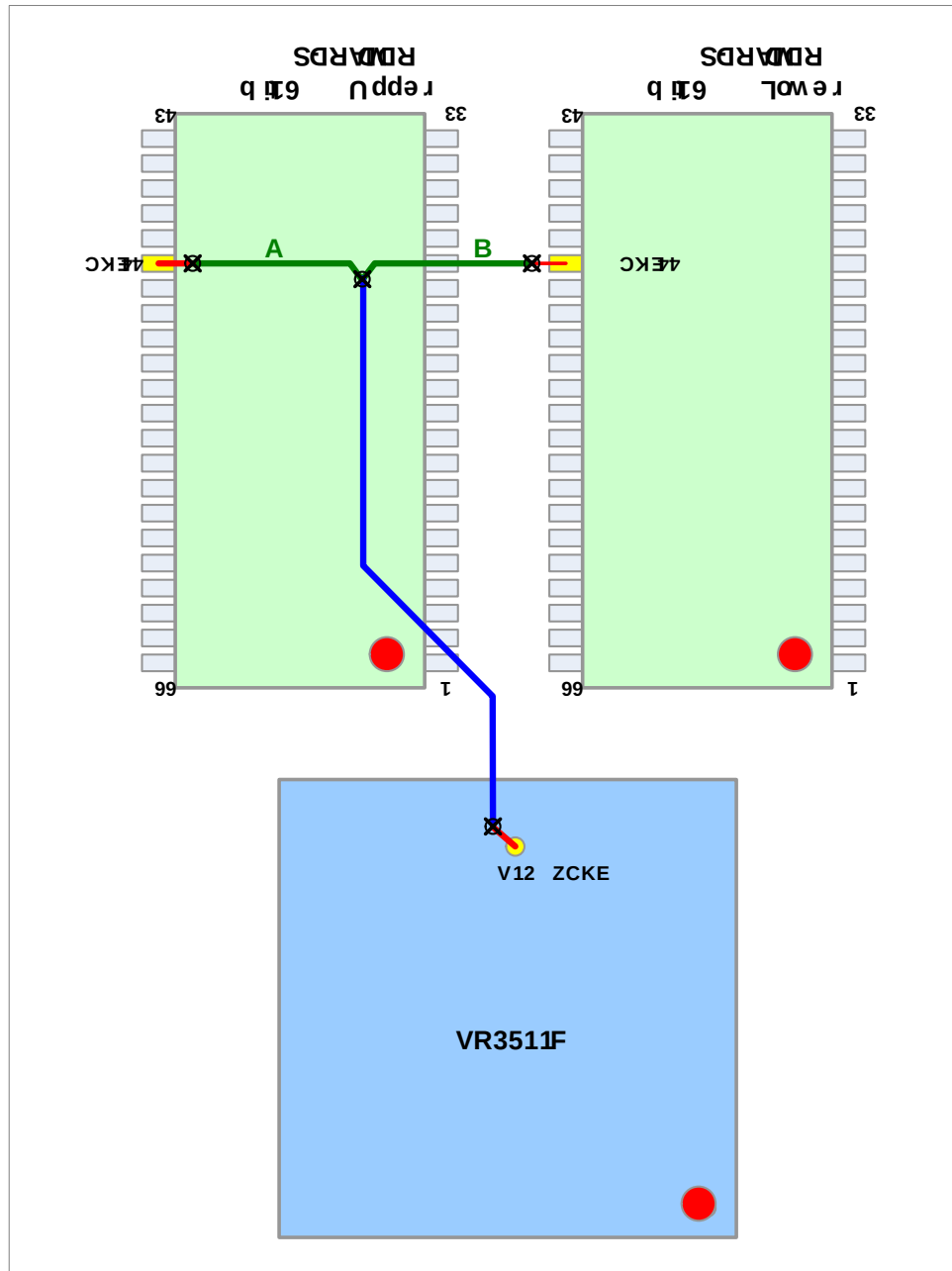
Signal Name	Length (mm)	Width (mm)	Via hole
DQ(Data bus)	40 ~ 55	0.15	0
DQS	55 ~ 65	0.30	2
CLK + iCLK	55 ~ 65	0.30	3
nCLK + inCLK	55 ~ 65	0.30	3
DM	40 ~ 55	0.30	2
CKE	40 ~ 55	0.15	3
nCS	40 ~ 55	0.15	3
nRAS	40 ~ 55	0.15	3
nCAS	40 ~ 55	0.15	3
nWE	40 ~ 55	0.15	3
Address	40 ~ 55	0.15	3

<Table 1. Recommended values for each signal's length and width>

In case of 32bit DDR-SDRAM data bus system, two DDR-SDRAMs which has 16bit data bus width respectively are necessary. In this case, unlike DQ(Data bus), DQS, and DM signal, the rest signals(CLK, nCLK, CKE, nCS, nRAS, nCAS, nWE, and Address) are connected to both DDR-SDRAM and these signals need three via hole. On the other hand, DQ, DQS, and DM signals are connected to only one DDR-SDRAM and need two via hole.

<Fig 8> shows the pattern routing rule for three via hole case. GREEN and BLUE-colored

signal are inner-layer ZCKE signal in PCB. RED-colored signal is top(or bottom) plane ZCKE signal on PCB. Although both GREEN and BLUE signal are inner-layer signal, they must be located on separated-layer respectively. And the length of **A** and **B** must be equal.



- ⊗ : Via hole (or Through hole)
- : ZCKE signal on top(or bottom) plane of PCB
- : ZCKE signal in innerlayer of PCB
- : ZCKE signal in innerlayer of PCB

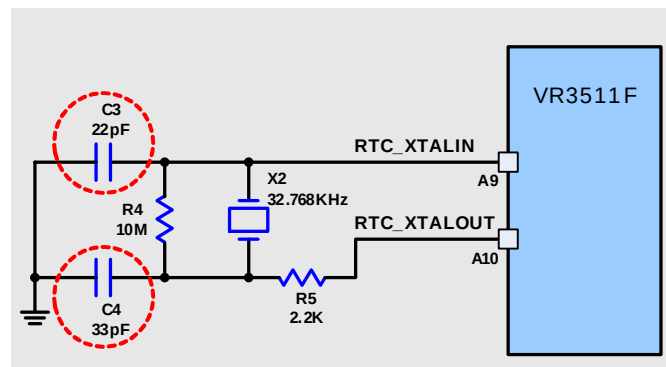
<Fig 8. Pattern routing rule for 3 via hole signal case>

2. Design guide for Clock source

10MHz and 32.768KHz Crystal are basic clock source of the VR3511F. 10MHz clock signal is an input for internal PLL and 32.768KHz clock signal is an input for RTC and used to make control signals (Static Bus control signals, Reset output signal, and so on)

2.1. Circuit design guide

The customer of VR3511F has to pay attention to the load capacitor selection of RTC crystal. <Fig 9> shows the detail circuit of RTC clock part of VR3511F Navigation Development Board. C3 and C4 value are 22pF and 33pF respectively. However it is not an absolute value because load capacitor value is dependent on crystal vendor. 22pF & 33pF are just a value for VR3511F Navigation Development Board case. Do not think these values are adaptable to your own VR3511F product. For stable oscillation of RTC crystal, H/W developer must check the load capacitor value that crystal vendor recommends.



<Fig 9. Load capacitor selection for RTC Crystal>

2.2. PCB layout design guide

This PCB Layout guide describes the crystal location, each clock signal's pattern width, length, via hole count. The customer of VR3511F must follow this bulletin's PCB Layout guide.

2.2.1. PCB layout rule related to clock source

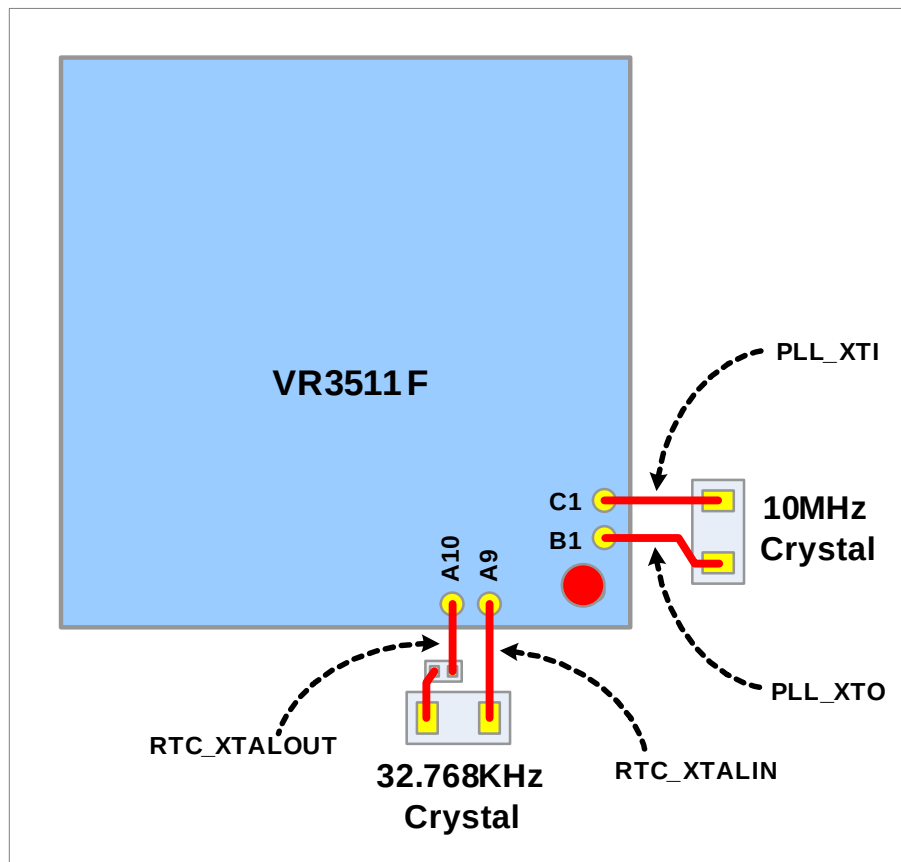
- 01) VR3511F, PLL crystal, and RTC crystal must be located on same plane of PCB
- 02) The location of RTC and PLL Crystal must be same as <Fig 7>
- 03) The width of PLL clock signal line must be thicker than the width of normal signal line
- 04) The width of RTC clock signal line must be thicker than the width of normal signal line

- 05) Do not use via hole(through hole) on PLL clock signal line
- 06) Do not use via hole(through hole) on RTC clock signal line
- 07) Minimize the length between VR3511F and PLL crystal as close as possible
- 08) Minimize the length between VR3511F and RTC crystal as close as possible

Recommended values for each clock signal's length and width are as follows. The customer of VR3511F must follow the rule of <Table 2> and <Fig 10>.

Signal Name	Length (mm)	Width (mm)	Via hole
PLL_XTI	~ 12	0.30	0
PLL_XTO	~ 12	0.30	0
RTC_XTALIN	~ 12	0.30	0
RTC_XTALOUT	~ 12	0.30	0

<Table 2. Recommended values for each signal's length and width>



<Fig 10. Pattern routing rule for clock signal>